Engineering Chemistry (AHT-002) - Unit 5

Unit 5 - Spectroscopy, Organic Reactions & Drug Synthesis - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): spectroscopy (principle, selection rules, electronic/vibrational/rotational), NMR and MRI, organic reaction types (addition, elimination, substitution, cyclization - Diels-Alder), and synthesis of one drug (aspirin, phenacetin, melubrin, novalgin).

Key ideas: Spectroscopy studies how molecules absorb electromagnetic radiation: rotational (microwave), vibrational (infrared), electronic (UV-visible) - energy gaps decide the region. **Selection rules** say which transitions are allowed (vibrational needs a changing dipole moment; rotational needs a permanent dipole). IR is used to identify functional groups from characteristic stretching frequencies. **NMR:** nuclei with spin (^{1}H) flip between spin states in a magnetic field; chemical shift tells the proton's environment, splitting (n+1 rule) tells the neighbours. **MRI** applies the same physics to image body tissue - see the simple notes under this unit. **Organic reactions:** addition (across double bonds), elimination (form double bonds), substitution (swap a group), and the **Diels-Alder** cycloaddition (diene + dienophile → six-membered ring). **Aspirin synthesis:** salicylic acid + acetic anhydride (esterification of the -OH group).

Exam pointers: "principle of NMR spectroscopy" and "application of IR" are the common theory questions; one Diels-Alder mechanism and the aspirin synthesis (equation with conditions) appear regularly. Draw the MO-free, arrow-pushing steps cleanly - mechanism marks are step marks.

Extra note - MRI - made simple:

MRI (Magnetic Resonance Imaging) = a machine that photographs the inside of the body, working on the principle of **NMR spectroscopy**.

> *Just as NMR finds the structure of small molecules by applying a magnetic field, MRI maps the whole body (mostly the hydrogen atoms in water).*

- Hydrogen (in water) is everywhere in the body. A strong magnetic field + radio waves are applied.
- Hydrogen in each tissue gives a different signal - a computer turns that into an image.
- **Advantage:** no harmful radiation (unlike X-rays), and a clear image of soft tissue.

Exam tip: "MRI is an application of NMR" - do not forget to write this one line.